

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination 2024-25 (NEP)

DEPARTMENT OF GEOLOGY / EARTH SCIENCE**Paper Code: ESMJ11 / ESMN111 — Elements of Earth Sciences**

Time: 2:30 Hours

Maximum Marks: 70

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Attempt five questions in all, selecting at least two questions each from Section-A and Section-B. Section-C is compulsory. All questions carry equal marks (14 marks each).***SECTION – A****Question 1**

[14 Marks]

Write descriptive notes about Nebular Hypothesis and Protoplanet Hypothesis.

Question 2

[14 Marks]

What is the concept of Uniformitarianism? Also write explanatory notes on the concepts of Neptunism, Vulcanism and Plutonism.

Question 3

[14 Marks]

Differentiate between:

- (a) Continental crust and Oceanic crust
- (b) Mud-slide and Debris-slide
- (c) Relative dating and Absolute dating
- (d) Strike joint and Dip joint

SECTION – B**Question 4**

[14 Marks]

What are the causes of volcanic eruptions? With neat diagrams, discuss the classification of volcanoes.

Question 5

[14 Marks]

What is a fold? With neat diagram describe different parts of a fold. Also describe the classification of folds based on:

- (a) Upward and downward closing of folds
- (b) Interlimb angle of folds

Question 6

[14 Marks]

Write critical notes on any **two** of the following:

- (a) Propagation of P- and S-waves inside the Earth
- (b) Principle of Superposition and its use in field geology
- (c) Causes of earthquakes
- (d) Fault classification based on the rake of the net slip

SECTION – C (Compulsory)**Question 7**

[14 Marks]

Describe the following in two or three sentences each:

- (a) Mass of the Earth
- (b) Repetti discontinuity
- (c) NiFe
- (d) Catastrophism
- (e) Weathering
- (f) Rayleigh waves
- (g) Palaeozoic era
- (h) Half-life of ^{14}C
- (i) Caldera
- (j) True dip
- (k) Antiformal syncline
- (l) Throw of fault
- (m) Outcrop
- (n) Contour line